MULTIPLE LINEAR REGRESSION

A multiple regression is used to predict a continuous dependent variable based on multiple independent variables or interaction between multiple independent variables.

As such, it extends simple linear regression, which is used when you have only one continuous independent variable.

it will let you:

(a) determine whether the linear regression between a combination of independent variables (a linear regression model) and a continuous dependent variable is statistically significant;

(b) determine how much of the variation in the dependent variable is explained by the changes in the combination of independent variables;

(c) understand the direction and magnitude of linear relationship between the dependent variable and a group of independent variables; and

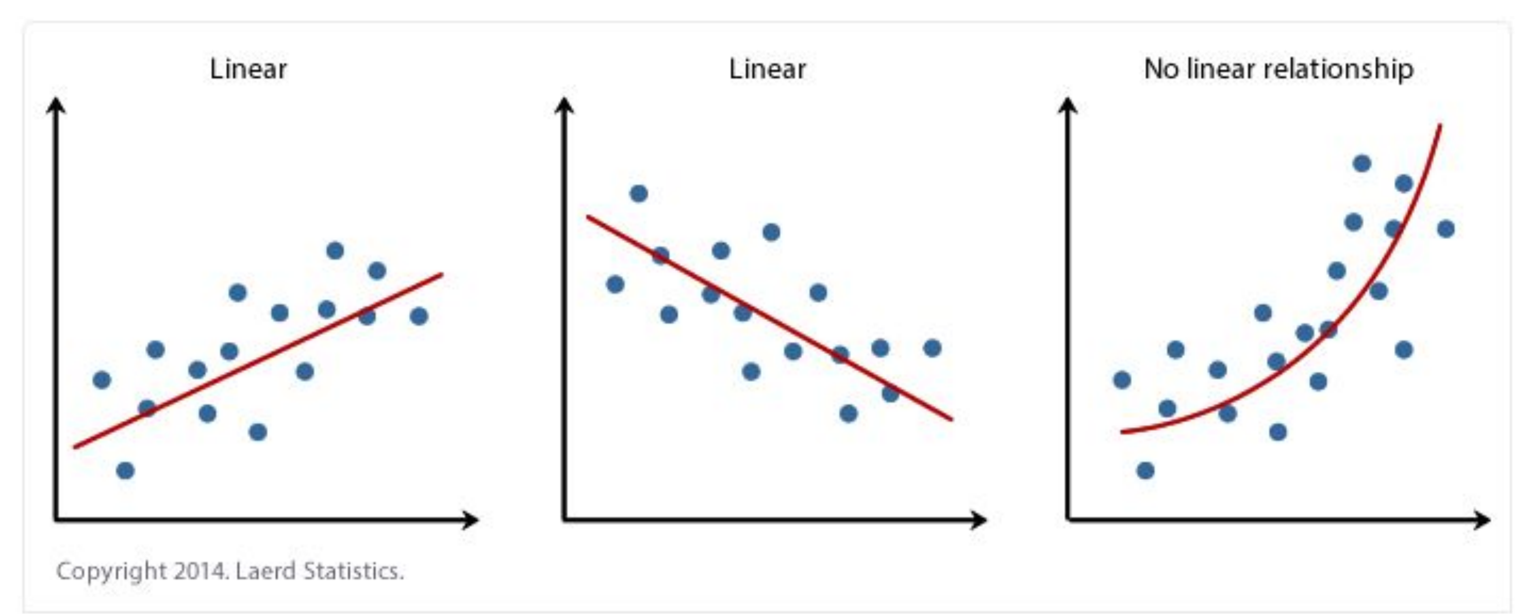
(d) predict values of the dependent variable based on different values of the independent variables.

6 assumptions must be satisfied for doing a linear regression analysis

1. **you have a continuous dependent variable;**

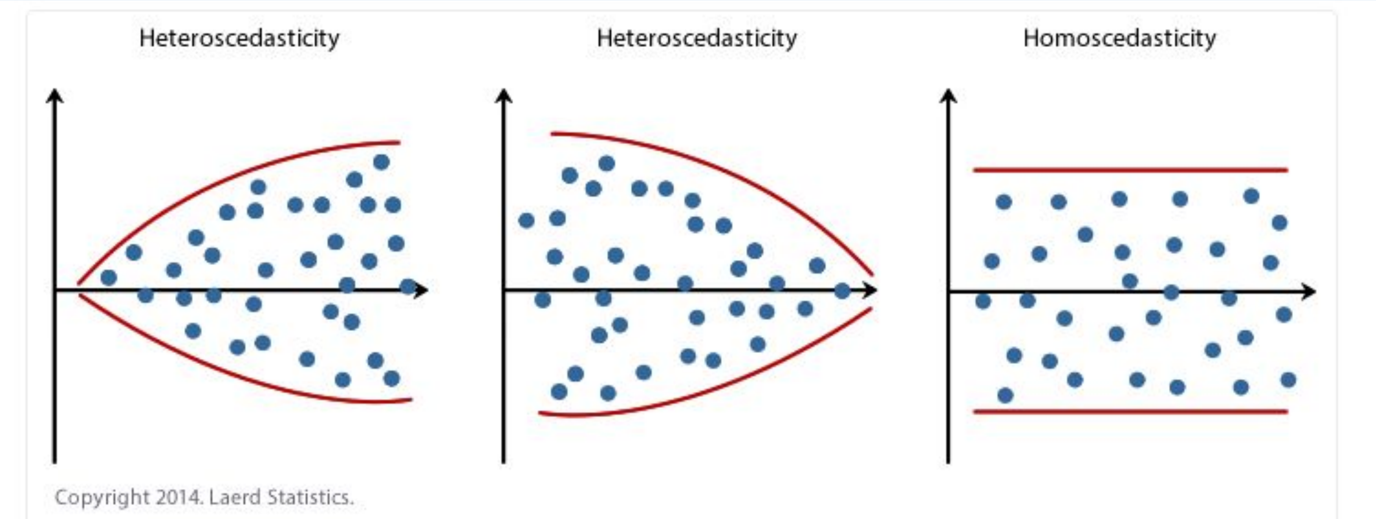
2. **you have continuous or categorical independent variable.**

3. **there is a linear relationship between the dependent variable and independent variables;**

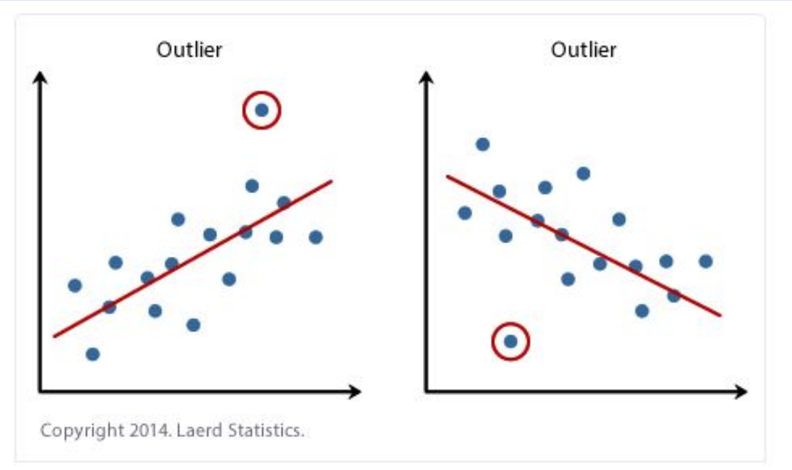


4. **you have independence of observations; - Durbin Watson Statistics should be near to 2 – shows non-autocorrelation (< 1 or >3 is not acceptable)**

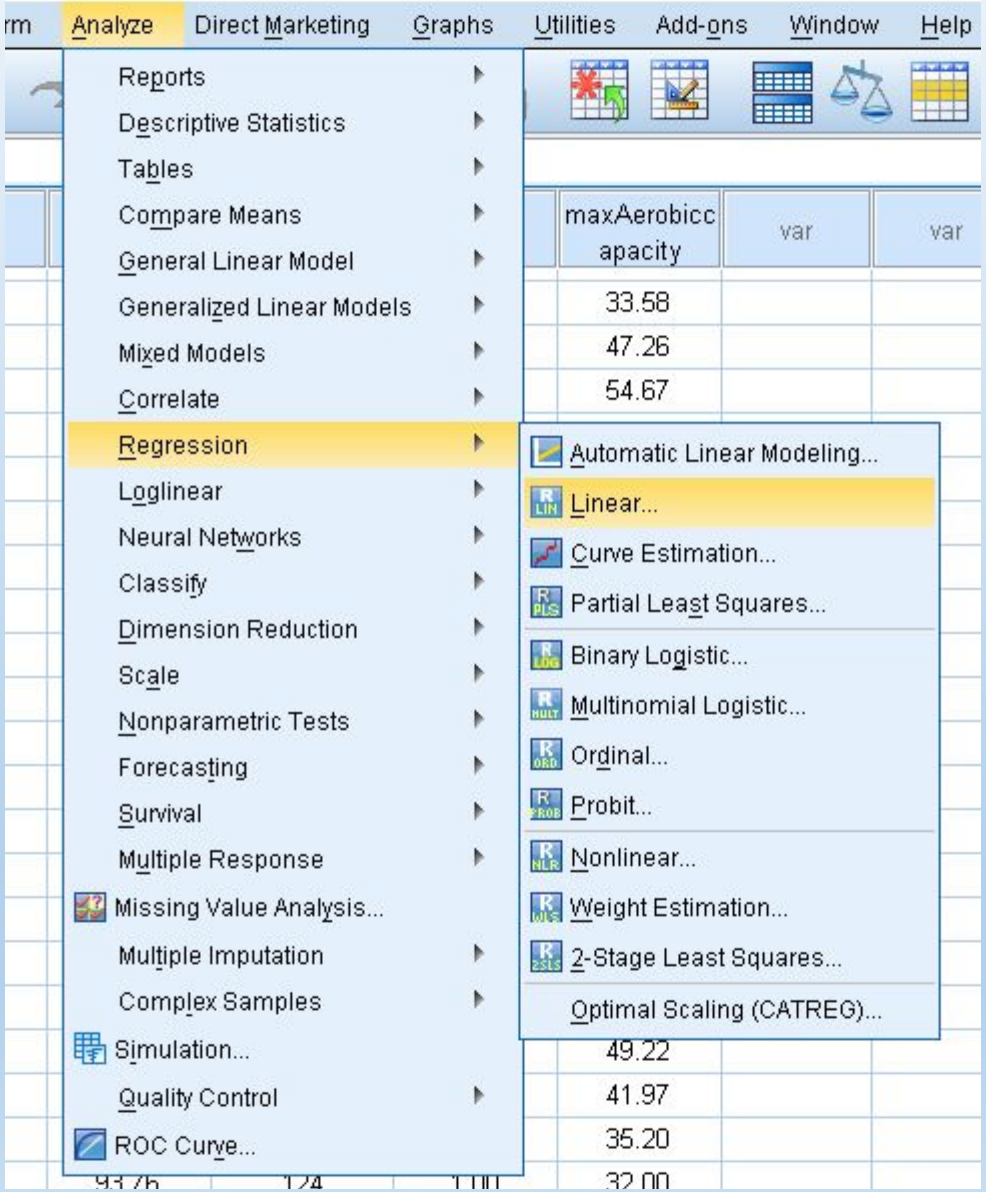
5. there is homoscedasticity;

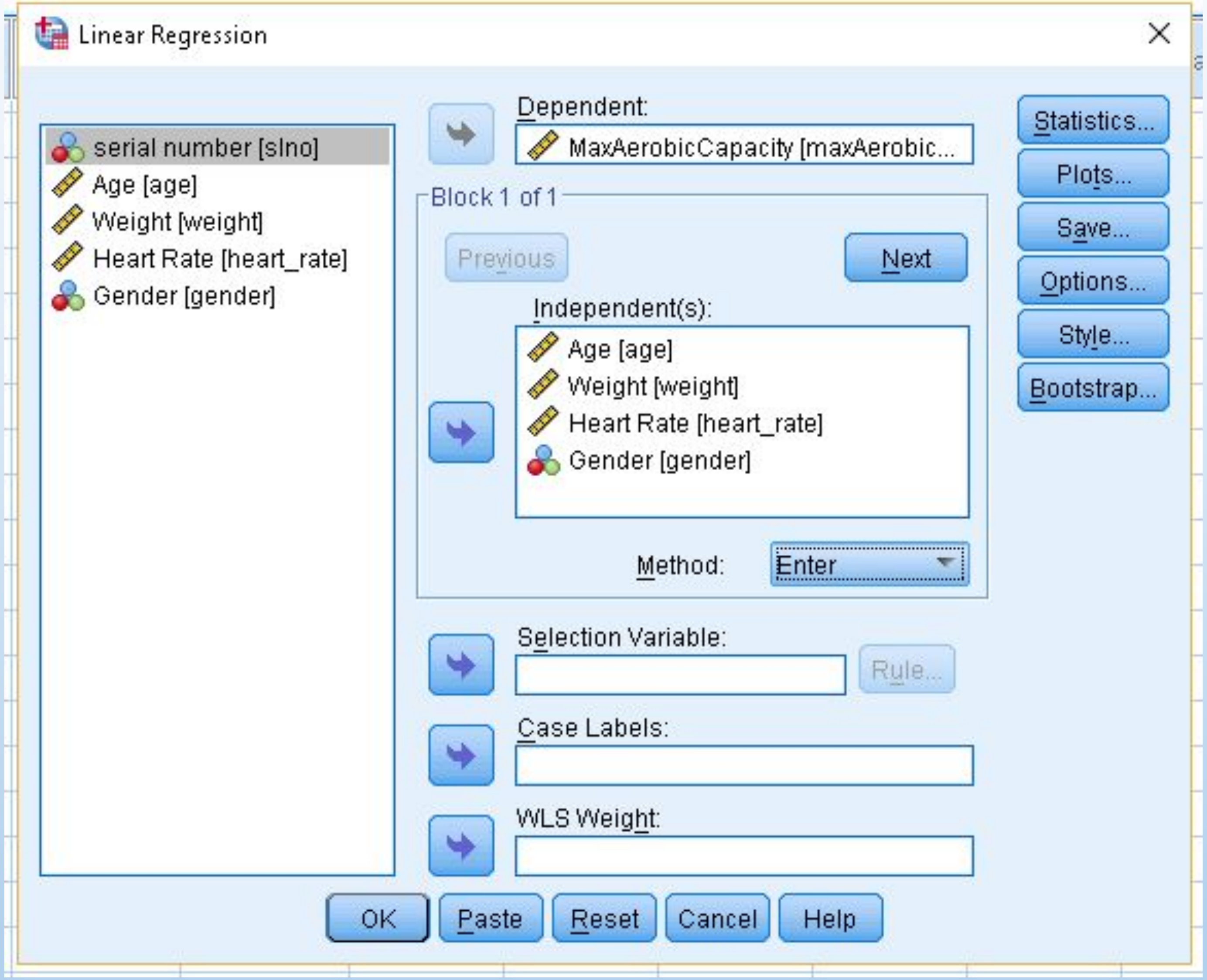


6. there are no significant outliers



Analyze -------> Regression ----------> Linear



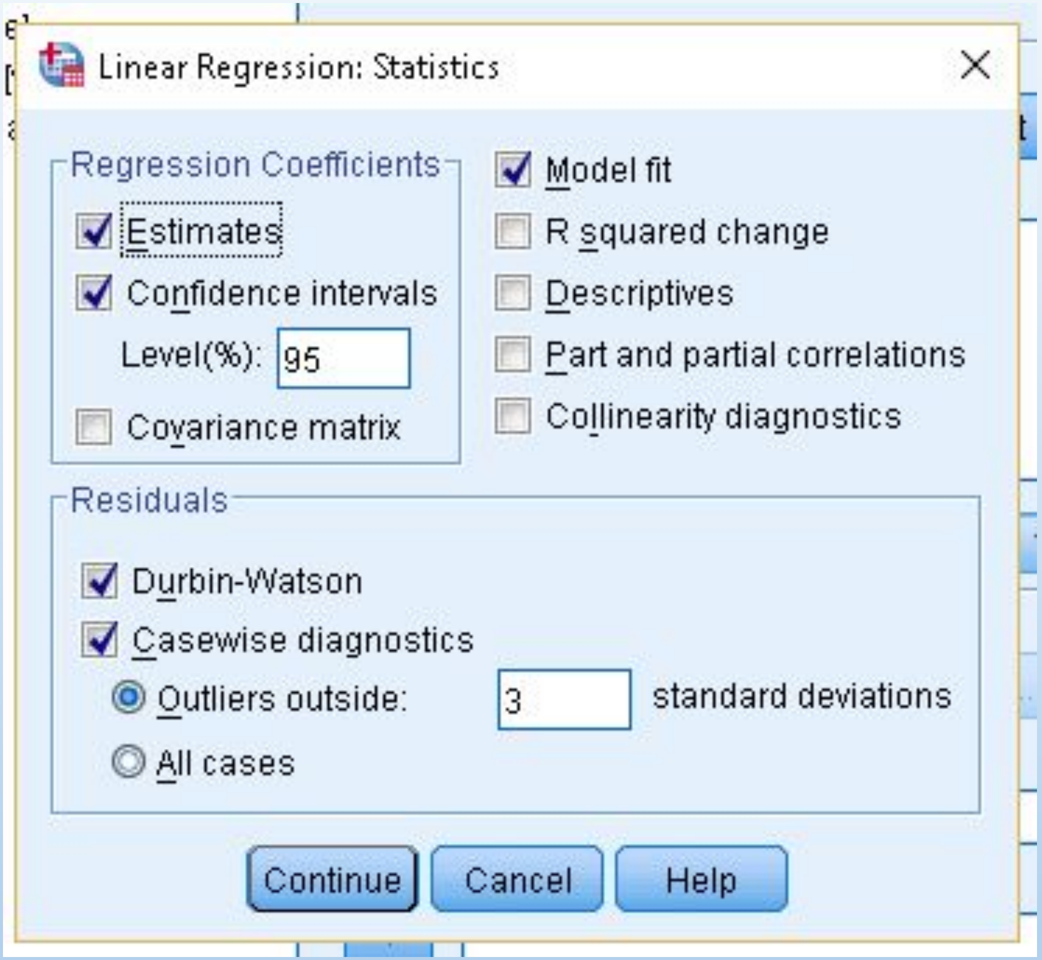


In our example, we need to enter the variable "maximum aerobic capacity" as the dependent variable and age, weight, heart rate, and gender as independent variables. The default method for the multiple linear regression analysis is 'Enter'. That means that all variables are forced to be in the model.

Click on Statistics

Check

- Estimates,
- Confidence intervals level (95%)
- Durbin Watson
- Casewise diagnostics – outliers outside 3 standard deviations
- Model fit



Click Continue -------------------> Then OK

Model Summary[b]

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate	Durbin-Watson
1	.760[a]	.577	.559	5.69097	1.910

a. Predictors: (Constant), Gender, Age, Heart Rate, Weight

b. Dependent Variable: MaxAerobicCapacity

ANOVA[a]

Model		Sum of Squares	df	Mean Square	F	Sig.
1	Regression	4196.483	4	1049.121	32.393	.000[b]
	Residual	3076.778	95	32.387		
	Total	7273.261	99			

a. Dependent Variable: MaxAerobicCapacity

b. Predictors: (Constant), Gender, Age, Heart Rate, Weight

Coefficients[a]

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.	95.0% Confidence Interval for B	
		B	Std. Error	Beta			Lower Bound	Upper Bound
1	(Constant)	114.246	7.506		15.220	.000	99.345	129.148
	Age	-.165	.063	-.176	-2.633	.010	-.290	-.041
	Weight	-.385	.043	-.677	-8.877	.000	-.471	-.299
	Heart Rate	-.118	.032	-.252	-3.667	.000	-.182	-.054
	Gender	-13.208	1.344	-.748	-9.824	.000	-15.877	-10.539

a. Dependent Variable: MaxAerobicCapacity

MAC = 114.246 – 0.165 * Age – 0.385 * Weight – 0.118 * Heart Rate – 13.208 * Gender